BASH SCRIPTING IN VLSI EXPERT-

[b23n_sarif_727@prac3 myfolder]$ touch file_100

[b23n_sarif_727@prac3 myfolder]$ ls

file_100

(new empty file will be created)

[b23n_sarif_727@prac3 folder1]$ pwd

/ve/po_home/b23n_sarif_727/myfolder/folder1

[b23n_sarif_727@prac3 myfolder]$ gedit file_200

[b23n_sarif_727@prac3 myfolder]$ vi file_200

(Direct open new empty file in white space formate)

[b23n_sarif_727@prac3 myfolder]$ vi file_200

[b23n_sarif_727@prac3 myfolder]$ cat file_200

sarif ahmad vlsi engineer

[b23n_sarif_727@prac3 myfolder]$ ls -l

total 568

drwxr-xr-x 4 b23n_sarif_727 po   60 May 4 2025 custom_compiler_work

-rw-r--r-- 1 b23n_sarif_727 po   44 Mar 26 08:08 file1

-rw-r--r-- 1 b23n_sarif_727 po   0 Apr 2 2025 file10

-rw-r--r-- 1 b23n_sarif_727 po   0 Mar 26 13:56 file_100

-rw-r--r-- 1 b23n_sarif_727 po   149 Mar 26 08:08 file1.txt

-rw-r--r-- 1 b23n_sarif_727 po   37 Mar 26 13:40 file2

-rw-r--r-- 1 b23n_sarif_727 po   26 Mar 26 13:58 file_200

-rwxr-xr-x 1 b23n_sarif_727 po   135 Mar 26 08:10 file2.txt

(get the detail informations about the file and directory)

[b23n_sarif_727@prac3 myfolder]$ ls -ltr

total 568

-rw-r--r-- 1 b23n_sarif_727 po 491520 Apr 2 2025 typescript

-rw-r--r-- 1 b23n_sarif_727 po 18915 Apr 2 2025 index.html

drwxr-xr-x 2 b23n_sarif_727 po   6 Apr 2 2025 folder2

-rw-r--r-- 1 b23n_sarif_727 po   217 Apr 2 2025 files.tar.gz

-rw-r--r-- 1 b23n_sarif_727 po   46 Apr 2 2025 xab

-rw-r--r-- 1 b23n_sarif_727 po   44 Apr 2 2025 xaa



[b23n_sarif_727@prac3 myfolder]$ vi file1.txt

[b23n_sarif_727@prac3 myfolder]$ cat file1.txt

line1 is sarif

nine2 is ahmad

sine3 is vlsi

pine4 is expert

line5 is noida

line6 is delhi

line7 is kanpu

line is ashish

line is mohd

line is izhar

Searching string in a file

[b23n_sarif_727@prac3 myfolder]$ vi file2.txt

[b23n_sarif_727@prac3 myfolder]$ vi file1.txt



Vi filename then search after the specific pattern (:/sarif) then enter the press buttom

:/sarif



:noh (use this command to remove highlight string inside the file)

[b23n_sarif_727@prac3 myfolder]$ pwd

/ve/po_home/b23n_sarif_727/myfolder

[b23n_sarif_727@prac3 myfolder]$ head -5 file1.txt

line1 is sarif

nine2 is ahmad

sine3 is vlsi

pine4 is expert

line5 is Noida

[b23n_sarif_727@prac3 myfolder]$ tail -5 file1.txt

line7 is kanpu

line is ashish

line is mohd

line is izhar



[b23n_sarif_727@prac3 myfolder]$ wc -w xaa

9 xaa

[b23n_sarif_727@prac3 myfolder]$ wc -l xaa

3 xaa

Searching file:

[b23n_sarif_727@prac3 myfolder]$ find file*

file1

file10

file_100

file1.txt

file2

file_200

file2.txt

[b23n_sarif_727@prac3 myfolder]$ find *.txt
file1.txt
file2.txt
newfile1.txt

[b23n_sarif_727@prac3 myfolder]$ ls file*
file1 file_100 file2 file2.txt file3.sh file4.csv file5.v file7 file9
file10 file1.txt file_200 file3 file4 file5 file6 file8 files.tar.gz

[b23n_sarif_727@prac3 folder1]$ touch file{1..10}
ls
file1 file2 file3 file4 file5 file6 file7 file8 file9 file10

[b23n_sarif_727@prac3 myfolder]$ shuf file1
my name is sarif ahmad.
hellow how are you.

[b23n_sarif_727@prac3 myfolder]$ cmp file1 file2
file1 file2 differ: byte 1, line 1

[b23n_sarif_727@prac3 myfolder]$ cmp file1 file2
file1 file2 differ: byte 1, line 1
[b23n_sarif_727@prac3 myfolder]$ diff file1 file2
0a1
> hello users

```
2c3
< hellow how are you.
---

[b23n_sarif_727@prac3 myfolder]$ vi file1.txt
[b23n_sarif_727@prac3 myfolder]$ cat file1.txt | grep sarif
line1 is sarif

[b23n_sarif_727@prac3 myfolder]$ ps
    PID TTY          TIME CMD
 255357 pts/7    00:00:00 bash
 462992 pts/7    00:00:00 ps

[b23n_sarif_727@prac3 myfolder]$ which vcs
/usr/synopsys/vcs/W-2024.09-1/bin/vcs

[b23n_sarif_727@prac3 myfolder]$ cal 2024
                              2024

      January               February                March
Su Mo Tu We Th Fr Sa  Su Mo Tu We Th Fr Sa  Su Mo Tu We Th Fr Sa
    1  2  3  4  5  6               1  2  3                  1  2
 7  8  9 10 11 12 13   4  5  6  7  8  9 10   3  4  5  6  7  8  9
14 15 16 17 18 19 20  11 12 13 14 15 16 17  10 11 12 13 14 15 16
21 22 23 24 25 26 27  18 19 20 21 22 23 24  17 18 19 20 21 22 23
28 29 30 31           25 26 27 28 29        24 25 26 27 28 29 30
                                            31
```

```
      April                May                June
Su Mo Tu We Th Fr Sa  Su Mo Tu We Th Fr Sa  Su Mo Tu We Th Fr Sa
   1  2  3  4  5  6           1  2  3  4                  1
 7  8  9 10 11 12 13   5  6  7  8  9 10 11   2  3  4  5  6  7  8
14 15 16 17 18 19 20  12 13 14 15 16 17 18   9 10 11 12 13 14 15
21 22 23 24 25 26 27  19 20 21 22 23 24 25  16 17 18 19 20 21 22
28 29 30              26 27 28 29 30 31     23 24 25 26 27 28 29
                                            30
```

[b23n_sarif_727@prac3 myfolder]$ uptime

15:11:39 up 16:04, 5 users, load average: 0.14, 0.49, 0.82



[b23n_sarif_727@prac3 myfolder]$ vi newfile1.txt

[b23n_sarif_727@prac3 myfolder]$ awk '{print $1}' newfile1.txt

sarif

shaya

ashish

izhar

talha

[b23n_sarif_727@prac3 myfolder]$ awk '{print}' newfile1.txt

sarif 24 ece

shaya 25 ee

ashish 23 cmp

izhar 22 math

talha 27 physics

[b23n_sarif_727@prac3 myfolder]$ awk '{print $1, $3}' newfile1.txt

sarif ece

shaya ee

ashish cmp

izhar math

talha physics

[b23n_sarif_727@prac3 myfolder]$ cat file1.txt file2.txt > comb_file.txt

[b23n_sarif_727@prac3 myfolder]$ ls

comb_file.txt

```
line1 is sarif
nine2 is ahmad
sine3 is vlsi
pine4 is expert
line5 is noida
line6 is delhi
line7 is kanpu
line is ashish
line is mohd
line is izhar
r
line1 is first
line2 is second
line3 is third
line4 is four
line5 is five
line3 is third
line1 is first
line2 is second
line3 is third
```

[b23n_sarif_727@prac3 myfolder]$ split -l 2 file1.txt

[b23n_sarif_727@prac3 myfolder]$ ls

Xae xaf

[b23n_sarif_727@prac3 myfolder]$ mkdir folder{4..5}

[b23n_sarif_727@prac3 myfolder]$ ls

folder4 folder5

[b23n_sarif_727@prac3 myfolder]$ cat file1.txt | wc -l

11

[b23n_sarif_727@prac3 myfolder]$ cat file1.txt | wc -w

31

[b23n_sarif_727@prac3 mybash_practice]$ which bash

/usr/bin/bash (path where tool is installed)

#!/bin/bash → command line on top in the file (**filename.sh**) after opening vi editor

[b23n_sarif_727@prac3 mybash_practice]$ echo helloworld

helloworld

R W X (7 7 7)

1 1 1

chmod 777 mybash.sh

ls -ltr

[b23n_sarif_727@prac3 mybash_practice]$ echo "helloworld"

helloworld

[b23n_sarif_727@prac3 mybash_practice]$ put "helloworld"

bash: put: command not found...

in TCL → enter keyword **tclsh**

[b23n_sarif_727@prac3 mybash_practice]$ tclsh

% put "helloworld"

helloworld

% echo "helloworld"

helloworld

% exit

[b23n_sarif_727@prac3 mybash_practice]$

[b23n_sarif_727@prac3 mybash_practice]$ vi mybash_1.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./mybash_1.sh

bash: ./mybash_1.sh: Permission denied

[b23n_sarif_727@prac3 mybash_practice]$ chmod 777 mybash_1.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./mybash_1.sh

helloworld

in mybash_1.sh file data was present:



#!/bin/bash

echo "helloworld"

```
#!/bin/bash

name="sarif ahmad anjanay ritik abhishek aman sandhya shubham shukla nitimam vyomsir"
echo "Hello $name"
~
```

[b23n_sarif_727@prac3 mybash_practice]$ vi mybash_1.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./mybash_1.sh

Hello sarif ahmad anjanay ritik abhishek aman sandhya shubham shukla nitimam vyomsir

```
#!/bin/bash

name="sarif ahmad anjanay ritik abhishek aman sandhya shubham shukla nitimam vyomsir"
echo -e "Hello ${name// /\\nHello} "
```

[b23n_sarif_727@prac3 mybash_practice]$ vi mybash_1.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./mybash_1.sh

Hello sarif

Hello ahmad

Hello anjanay

Hello ritik

Hello abhishek

Hello aman

Hello sandhya

Hello shubham

Hello shukla

Hello nitimam

Hello vyomsir

For neglecting the line in the file (**filename.sh**) use command → #sarif ahmad

External command line: at run time

```
#!/bin/bash

echo "Enter the name :"
read name
echo "Hello, $name"
```

[b23n_sarif_727@prac3 mybash_practice]$ vi mybash_1.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./mybash_1.sh

Enter the name :

SARIF_AHMAD

Hello, SARIF_AHMAD

```
#!/bin/bash
a=5
b=2
sum=$((a+b))
diff=$((a-b))
mul=$((a*b))
div=$((a/b))
rem=$((a%b))
square1=$((a*a))
square2=$((b*b))
power=$((a**b))
echo "$sum,$diff,$mul,$div,$rem,$square1,$square2,$power"
~
```

[b23n_sarif_727@prac3 mybash_practice]$ vi file1.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./file1.sh

7,3,10,2,1,25,4,25

```
#!/bin/bash

echo "What is your name?"
read name
echo "Hi there $name"
echo "Welcome to ahmad!"

~
```

[b23n_sarif_727@prac3 mybash_practice]$ vi pract_file.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./pract_file.sh

What is your name?

sarif

Hi there sarif

Welcome to ahmad!

```
#!/bin/bash

read -p "What is your name? " name
echo "Hi there $name"
echo "Welcome to ahmad!"
```

[b23n_sarif_727@prac3 mybash_practice]$ vi file.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./file.sh

What is your name? sarif

Hi there sarif

Welcome to ahmad!

By user at run time:

```
#!/bin/bash
echo "number1"
read number1
echo "number2"
read number2

sum=$((number1+number2))
echo "sum of two nums is: $sum"
~
```

[b23n_sarif_727@prac3 mybash_practice]$ vi file2.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./file2.sh

number1

50

number2

10

sum of two nums is: 60

Printing an even number: at run time

```
#!/bin/bash
echo "enter a number : "
read number

if (($number % 2== 0));
then
        echo "Number is even"

else
        echo "Number is odd"


fi
```

[b23n_sarif_727@prac3 mybash_practice]$ vi file3.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./file3.sh

enter a number :

4

Number is even

```
#!/bin/bash

echo "enter a mount point"
read mount_point

if [ "$(mount | grep -w $mount_point )" ];then
        echo "mount point is used"
else
        echo "mount point is not used"

fi
```

- ✓ mount → shows all mounted file systems
- ✓ grep -w $mount_point → **searches exact match**
- ✓ If found → condition TRUE
- ✓ If empty → condition FALSE

[b23n_sarif_727@prac3 mybash_practice]$ vi file4.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./file4.sh

enter a mount point

mount

mount point is not used

[b23n_sarif_727@prac3 mybash_practice]$ ./file4.sh

enter a mount point

sarif

mount point is not used

for loop

Enter a name & print a name:

```
#!/bin/bash

echo "enter a name:"
read name

for name in sarif shubham ritik anjay;
do
        if  (( $name == "sarif|| shubham ||ritik || anjay" ));
        then

                echo "Hii $name"
                break

        fi

done
```

[b23n_sarif_727@prac3 mybash_practice]$ vi for_file5.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./for_file5.sh

enter a name:

sarif

Hii sarif

Array: [@]

- **${arr[@]} → prints all elements of the array**
- **@ → means “all values”**

```
#!/bin/bash

mylist=("sarif" "ritik" "anjay" "shubham")
echo "enter a name"
read input

for item in "${mylist[@]}"; do
        if [ "$input"  = "$item" ]; then

                echo "Hii you entered a valid name: $input"

                break

        fi

done
```

[b23n_sarif_727@prac3 mybash_practice]$ vi file6.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./file6.sh

enter a name

sarif

Hii you entered a valid name: sarif

[b23n_sarif_727@prac3 mybash_practice]$ ./file6.sh

enter a name

altaf



```
#!/bin/bash

var="sarif"
echo "my name is $var"
echo 'my name is $var'

echo "$(date)"
```

[b23n_sarif_727@prac3 mybash_practice]$ vi file7.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./file7.sh

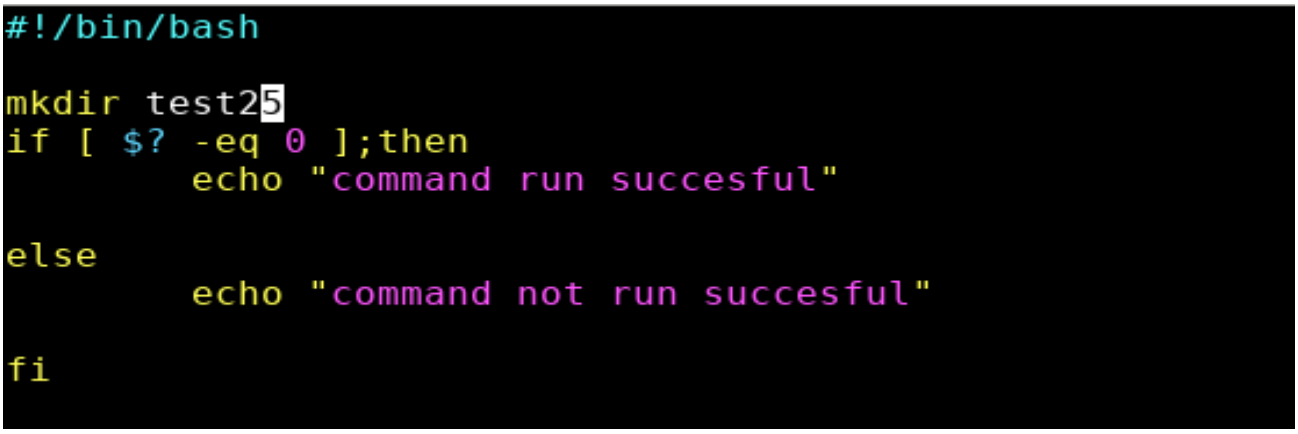
my name is sarif

my name is $var

Thu Mar 26 09:52:03 IST 2026

[b23n_sarif_727@prac3 mybash_practice]$ $?

bash: 0: command not found...



```
#!/bin/bash

mkdir test25
if [ $? -eq 0 ];then
        echo "command run succesful"

else
        echo "command not run succesful"

fi
```

[b23n_sarif_727@prac3 mybash_practice]$ vi file8.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./file8.sh

command run succesful

[b23n_sarif_727@prac3 mybash_practice]$ ls

5 file13.sh file16.sh file1.sh file4.sh file8.sh for_file5.sh output.txt **test25**

[b23n_sarif_727@prac3 mybash_practice]$ cd test25

[b23n_sarif_727@prac3 test25]$ ls

[b23n_sarif_727@prac3 mybash_practice]$ mkdir file{1,2,3}.txt

[b23n_sarif_727@prac3 mybash_practice]$ ls

file1.txt file2.txt file3.txt

[b23n_sarif_727@prac3 mybash_practice]$ echo "XYZ"; echo "ABC"

XYZ

ABC

```
#!/bin/bash

arr=(10 20 30 40)
echo "print all array element: ${arr[@]}"

arr1=(XYZ PQR ABC)
echo "print all array element: ${arr1[@]}"
```

[b23n_sarif_727@prac3 mybash_practice]$ vi mybash_1.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./mybash_1.sh

print all array element: 10 20 30 40

print all array element: XYZ PQR ABC

```
#!/bin/bash

arr=(10 20 30 40)
echo "print second array element: ${arr[1]}"
```

[b23n_sarif_727@prac3 mybash_practice]$ ./mybash_1.sh

print second array element: 20

```
#!/bin/bash

str="VYOM"
echo "${#str}"

echo {1..5}
```

[b23n_sarif_727@prac3 mybash_practice]$ vi mybash_1.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./mybash_1.sh

4

1 2 3 4 5

[b23n_sarif_727@prac3 mybash_practice]$ echo "Hello, World" > output_file.txt

output_file.txt

Hello, World

[b23n_sarif_727@prac3 mybash_practice]$ echo "Sarif ahmad" >> output_file.txt

[b23n_sarif_727@prac3 mybash_practice]$ vi output_file.txt

Hello, World

Sarif ahmad

[b23n_sarif_727@prac3 mybash_practice]$ cat < output_file.txt

Hello, World

Sarif ahmad

Arithmetic evaluation

```
#!/bin/bash

a=5
b=7
sum=$(expr $a + $b)
echo "$sum"
~
```

[b23n_sarif_727@prac3 mybash_practice]$ vi file9.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./file9.sh

12

```
#!/bin/bash

echo "First argument: $1"
echo "Second argument: $2"
echo "Total arguments: $#"
~
```

[b23n_sarif_727@prac3 mybash_practice]$ vi mybash_1.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./mybash_1.sh Hello world

First argument: Hello

Second argument: world

Total arguments: 2

While loop

```
#!/bin/bash

x=10
while (( $x > 5 ));do
        echo "$x"
        x=$((x-1))

done
```

[b23n_sarif_727@prac3 mybash_practice]$ vi file10.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./file10.sh

10

9

8

7

6

[b23n_sarif_727@prac3 mybash_practice]$ vi file12.txt

```
sarif ahmad
aligarh
noida
delhi
agra
master
electronics
digital electronics
network theorm
for loop
sarif
```

[b23n_sarif_727@prac3 mybash_practice]$ vi file14.sh

```
#!/bin/bash

while read -r num;do
        echo "$num"

done < file12.txt
~
~
```

[b23n_sarif_727@prac3 mybash_practice]$ vi file14.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./file14.sh

sarif ahmad

aligarh

noida

delhi

agra

master

electronics

digital electronics

network theorm

for loop

sarif

Case statement:-

```
#!/bin/bash

echo "enter the number"
read num

case $num in
        1) echo "sarifahmad";;
        2) echo "masters in ece";;
        3) echo "noida agra";;
        4) echo "delhi hauzkhas";;
        *) echo "default case";;
esac
```

[b23n_sarif_727@prac3 mybash_practice]$ vi file13.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./file13.sh

enter the number

2

masters in ece

[b23n_sarif_727@prac3 mybash_practice]$ ./file13.sh

enter the number

10

default case

Function

```
#!/bin/bash

function function_name(){
        echo "VLSI Engineer Sarif"

}

function_name
~
```

[b23n_sarif_727@prac3 mybash_practice]$ vi file15.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./file15.sh

VLSI Engineer Sarif

```
#!/bin/bash

function function_name(){
        echo "$(($1 + $2))"

}

function_name 2 3
```

[b23n_sarif_727@prac3 mybash_practice]$ vi file16.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./file16.sh

5

```
#!/bin/bash

function function_name(){

        if (( $1 > 10 ));
        then
                return 1
        else
                return 0


fi

}

function_name 52

echo $?
~
```

[b23n_sarif_727@prac3 mybash_practice]$ vi file17.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./file17.sh

1

[b23n_sarif_727@prac3 mybash_practice]$ vi file18.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./file18.sh

sarif

```
#!/bin/bash

function function_name(){

        local var="sarif"
        echo $var


}
function_name
echo $var
```

eval

eval is used to execute a command that is stored as a string**.**

- cmd → contains a command as a string
- eval $cmd → executes that string as a real command

🧠 **Key Idea:**

- Without eval → it's just a string
- With eval → it becomes an executable command

```
#!/bin/bash

cmd="echo Hello Sarif"

eval $cmd
~
```

[b23n_sarif_727@prac3 mybash_practice]$ vi mybash_1.sh

[b23n_sarif_727@prac3 mybash_practice]$ ./mybash_1.sh

Hello Sarif